

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

53002051 - Mercados De Electricidad

DEGREE PROGRAMME

05BK - Máster Universitario En Ingeniería De La Energía

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	53002051 - Mercados de Electricidad
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	05BK - Máster Universitario en Ingeniería de la Energía
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Carlos Enrique Vazquez Martinez (Subject coordinator)	518	vazquez.martinez@upm.es	Tu - 11:00 - 13:00 W - 11:00 - 13:00 Th - 11:00 - 13:00 Please, request an appointment by email

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Economía De La Energía

3.2. Other recommended learning outcomes

The subject - other recommended learning outcomes, are not defined.

4. Skills and learning outcomes *

4.1. Skills to be learned

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CE13 - Entender la evolución y el funcionamiento de los mercados de petróleo, gas y electricidad. Conocer los principales tipos de diseño de los mercados de electricidad y gas que existen en la experiencia internacional y los criterios bajo los que se han diseñado, y ser capaz de analizar cuál es la regulación más adecuada para cada situación.

CE18 - Entender la optimización de costes en una empresa: coste marginal, coste medio, coste hundido, coste de oportunidad, aplicados al sector de la energía. Analizar costes en el sector de la energía.

CE4 - Comprender y aplicar los principios de funcionamiento, formación de precios y equilibrio en los mercados energéticos, tanto en condiciones de competencia perfecta como en condiciones de competencia imperfecta

CG1 - Aplicar conocimientos de ciencias y tecnologías avanzadas a la práctica profesional o investigadora de la Ingeniería Energética.

CG2 - Poseer capacidad para diseñar, desarrollar, implementar, gestionar y mejorar productos, sistemas y procesos en los distintos ámbitos energéticos, usando técnicas analíticas, computacionales o experimentales avanzadas.

CT1 - Aplica. Habilidad para aplicar conocimientos científicos, matemáticos y tecnológicos en sistemas relacionados con la práctica de la ingeniería.

CT10 - Conoce. Conocimiento de los temas contemporáneos.

CT11 - Usa herramientas. Habilidad para usar las técnicas, destrezas y herramientas ingenieriles modernas necesarias para la práctica de la ingeniería.

CT3 - Diseña. Habilidad para diseñar un sistema, componente o proceso que alcance los requisitos deseados teniendo en cuenta restricciones realistas tales como las económicas, medioambientales, sociales, políticas, éticas, de salud y seguridad, de fabricación y de sostenibilidad.

CT4 - Trabaja en equipo. Habilidad para trabajar en equipos multidisciplinares.

CT5 - Resuelve. Habilidad para identificar, formular y resolver problemas de ingeniería.

CT7 - Comunica. Habilidad para comunicar eficazmente.

CT9 - Se actualiza. Reconocimiento de la necesidad y la habilidad para comprometerse al aprendizaje continuo.

4.2. Learning outcomes

RA167 - Analizar y entender los diferentes tipos de diseño de los mercados de electricidad, tanto de energía como de servicios complementarios

RA136 - Conocer la situación actual, evolución hasta el momento y perspectivas futuras de los diferentes mercados de energía

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The course focuses on the study of electricity markets, both their regulatory design and the description of the economic characteristics of the sectors and their implications for the behavior of agents.

The full range of market periods will be addressed, from very short-term reserves to forward sales mechanisms operating several years in advance. The emphasis is not so much on the specific regulations operating in the different markets, but rather on their foundations, design criteria, and the alternatives that could have been applied at each point. In general, emphasis will be placed on an international analysis of the markets, analyzing different implementation alternatives that have been adopted around the world.

5.2. Syllabus

1. Introduction
2. Energy markets
 - 2.1. Market design: day-ahead and intraday markets
 - 2.2. Bidding models
3. Forward markets and retailing
4. Congestion management and losses
5. Operational reserves

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction Duration: 02:00 Lecture			
2	Follow-up test Duration: 00:05 Additional activities Energy markets: day-ahead and intra-day Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
3	Follow-up test Duration: 00:05 Additional activities Energy markets: day-ahead and intra-day Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
4	Follow-up test Duration: 00:05 Additional activities Energy markets: day-ahead and intra-day Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
5	Follow-up test Duration: 00:05 Additional activities Energy markets: bidding models Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
6	Follow-up test Duration: 00:05 Additional activities Long-term markets: retailing Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
7	Follow-up test Duration: 00:05 Additional activities Long-term markets: forward markets Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05

8	Mid-term exam Duration: 00:50 Additional activities Losses Duration: 01:10 Lecture			Mid-term exam Written test Progressive assessment Presential Duration: 00:50
9	Follow-up test Duration: 00:05 Additional activities Congestion management Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
10	Follow-up test Duration: 00:05 Additional activities Congestion management Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
11	Follow-up test Duration: 00:05 Additional activities Congestion management Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
12	Follow-up test Duration: 00:05 Additional activities Operational reserves Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
13	Follow-up test Duration: 00:05 Additional activities Operational reserves Duration: 01:55 Lecture			follow-up test Written test Progressive assessment Presential Duration: 00:05
14	Operational reserves Duration: 01:10 Lecture Mid-term exam Duration: 00:50 Additional activities			Mid-term exam Written test Progressive assessment Presential Duration: 00:50
15				
16				
17				Final exam Written test Progressive assessment Presential Duration: 02:00 Final exam Written test Global examination Presential

			Duration: 02:00
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Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
2	follow-up test	Written test	Face-to-face	00:05	2.28%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
3	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
4	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11

							CE4 CE13 CE18
5	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
6	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
7	follow-up test	Written test	Face-to-face	00:05	2.28%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
8	Mid-term exam	Written test	Face-to-face	00:50	37.5%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5

							CT7 CT9 CT10 CT11 CE4 CE13 CE18
9	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
10	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
11	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18

12	follow-up test	Written test	Face-to-face	00:05	2.27%	0 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
13	follow-up test	Written test	Face-to-face	00:05	2.28%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
14	Mid-term exam	Written test	Face-to-face	00:50	37.5%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18
17	Final exam	Written test	Face-to-face	02:00	75%	5 / 10	CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10

							CT11 CE4 CE13 CE18
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7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final exam	Written test	Face-to-face	02:00	100%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10 CT11 CE4 CE13 CE18

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Extraordinary exam	Written test	Face-to-face	02:00	100%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT4 CT5 CT7 CT9 CT10

						CT11
						CE4
						CE13
						CE18

7.2. Assessment criteria

For students taking the course through progressive assessment,

- 25% of the final grade corresponds to the follow-up tests. There is no minimum grade for each individual test, but an average grade of 4 is required. Also, it is required to take at least 70% of the follow-up tests. Otherwise, the student will have to take the whole final exam.
- 75% of the final grade corresponds to the two mid-term exams, each of them weighting 37.5%. A minimum grade of 5 is required in each of the midterm exams. In case any of the midterm exams fails under that minimum grade, the topics corresponding to that exam will have to be evaluated again at the final examination. The topics covered by midterm exams with grades over 5 will not have to be evaluated again at the final exam. The relevant grade for calculating the global average will be the last exam taken, for each part.

For students taking the course under the "final exam only" option,

- 100% of their final grade will be awarded for the final exam

When an extraordinary exam is required, the entire syllabus must be evaluated, and 100% of the grade will be awarded for the extraordinary exam

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
S. Stoft, "Power System Economics: Designing Markets for Electricity", 2002	Bibliography	Power markets, focus on market design
I.J. Pérez-Arriaga (Editor), "Regulation of the Power Sector", Springer, 2013	Bibliography	Power markets, focus on market design
S. Hunt, "Making Competition Work in Electricity", 2002	Bibliography	Power markets, focus on competition

9. Other information

9.1. Other information about the subject

Students are encouraged to take this course simultaneously with the Environmental And Renewable Energy Markets course, as both are related and topics are interlinked.

This course is related to Sustainable Development Goal number 7: affordable and clean energy